

# Applying Process Mining to Radiological Workflows: A Clinical Case Study

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**Abstract.** Process mining has gained prominence in healthcare because it provides data-driven methods for assessing and improving compliance with established procedures. Nevertheless, the domain also presents distinctive characteristics and exposes several open challenges. This study applies process mining to radiological workflows in a large university hospital, with two complementary goals. First, to identify improvement opportunities. We identified critical bottlenecks and significant temporal deviations, and also demonstrated low compliance with the standard reference process. Second, the project served as an empirical validation of how distinguishing characteristics and challenges manifest in this context and how they provide guidance for future work. The findings demonstrate the capability of process mining to uncover such actionable insights and to drive evidence-based optimization of both operational and clinical quality in hospital radiology services.

**Keywords:** process mining, healthcare, radiological workflows, challenges

## 1 Introduction

In recent years, the healthcare sector has increasingly embraced technological innovations to address growing challenges related to resource efficiency and service quality. For example, the Integrating the Healthcare Enterprise (IHE) initiative<sup>3</sup> is a collaborative effort among healthcare professionals and industry stakeholders aimed at enhancing interoperability among healthcare information systems. IHE promotes the coordinated use of established standards such as DICOM [11] and HL7 [6] version 2.x for order and result management to meet specific clinical needs and support optimal patient care. The IHE initiative states that systems developed in accordance with its guidelines demonstrate improved interoperability, are easier to implement, and enable healthcare providers to leverage information more effectively, thereby helping to reduce operational costs and ensure high-quality care [9].

There is a growing interest in applying Process Mining [1] in the healthcare domain [10,13], known as Process Mining for Healthcare (PM4H). It enables

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<sup>3</sup> <https://www.ihe.net/>

data-driven analysis of clinical workflows and support compliance monitoring with established procedures. Radiology is one area where process mining applications are beginning to gain traction [7,5]. Within this context, the IHE initiative proposes the Scheduled Workflow (SWF) framework [9], which defines an integrated process for the ordering, scheduling, acquisition, storage, and visualization of radiological images. The core goal of the SWF profile is to significantly improve the efficiency and outcomes of radiological procedures, thereby enhancing the quality of care [9]. Process mining can play a key role in assessing and improving compliance with this framework, facilitating the identification of bottlenecks, inefficiencies, and deviations in clinical execution.

However, the successful application of process mining for healthcare (PM4H) requires addressing domain-specific challenges [10]. In this paper, we investigate the application of process mining to radiological workflows within the context of Hospital de Clínicas, a public university hospital complex in Uruguay that serves as a national reference. This case study is based on the analysis of event data extracted from the Health Information System (HIS), which is designed to interoperate with radiological devices using DICOM for image transfer and HL7 version 2.x for order and result messaging, and aligns with the SWF framework. We utilize patient attention times as a key quality indicator, as defined by the institution, to evaluate performance and identify process inefficiencies. The aim is twofold: to identify opportunities for improvement within the organization and to characterize the methodological challenges involved in applying process mining in a real-world healthcare domain.

The remainder of this paper is organized as follows. Section 2 describes related work. Section 3 presents the context in which the case study was performed, while Section 4 further describes the project. Section 5 discusses the experience in light of existing PM4H characteristics and challenges. Finally, Section 6 offers conclusions and outlines directions for future research.

## 2 Related work

In healthcare, process mining must cope with heterogeneous information systems, complex multidisciplinary pathways, and stringent ethical, privacy, and governance constraints. These characteristics and challenges have been comprehensively outlined in [10]. Evidence shows an uneven distribution of process mining applications across medical specialties. Recent work in radiation oncology demonstrates how process mining supports quality management and conformance checking, reinforcing its prominent role and offering transferable methodological insights [3]. Current research trends include: systematically combining process mining with process simulation to enable what-if analyses and informed decision making [14]; embedding process mining within continuous improvement frameworks (e.g., Kaizen) to institutionalize data-driven change [15]; and extending process mining to health equity monitoring, integrating fairness metrics into traditional analyses [2]. Additionally, recent case studies demonstrate the use

of end-to-end, cross-departmental pathway monitoring (encompassing radiology, laboratories, and other services) via advanced discovery techniques [12].

Despite its central diagnostic role and rich digital traceability, radiology remains comparatively underexplored in process mining research. Its operational complexity creates substantial opportunities to optimize end-to-end imaging workflows, implement urgency-aware prioritization, and integrate radiology events within broader hospital processes. Foundational contributions already point to crucial prerequisites: the adoption of standardized terminology to avoid ambiguity in event mapping [7], and the use of process mining to reveal heterogeneous PACS interaction patterns among radiologists [5]. The methodological advances highlighted above, supported by analysis [14] and a continuous-improvement-oriented process mining approach [15], outline a concrete roadmap to unlock this untapped potential.

### 3 Case Study context

Hospital de Clínicas operates a HIS with a distributed web-based architecture. The HIS integrates multiple healthcare subsystems (such as Admissions, Appointment Scheduling, Medical History, Emergency Room, and Study Requests) and non-healthcare subsystems that operate in a decentralized and interconnected manner. The Radiology Information System (RIS) can be considered integrated within the HIS in the Scheduling and Requests modules. The HIS is designed to seamlessly interoperate with radiological devices using DICOM for image management and HL7 version 2.x for the exchange of demographic data, orders and reports from patients, thus integrating the diagnostic modalities of different radiological device models. It also integrates standards and visual flows, as the imaging workflow is designed in compliance with the SWF profile, which is illustrated in Figure 1.

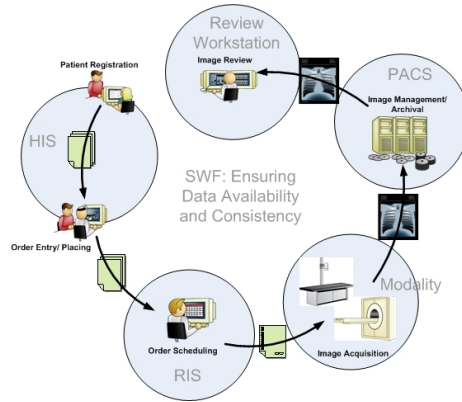


Fig. 1: SWF scheme (from [9])

In the Hospital de Clínicas, patient and radiological data flows from the Hospital Information System (HIS) to a Worklist server. For context, a Worklist is a critical interface that functions as a digital queue; it retrieves scheduled patients and their studies and sends this data directly to the imaging equipment, ensuring standardized and error-free transmission. This architecture is central to the SWF standard, which guarantees interoperability, security, and traceability. The process begins when a physician generates a study request in the HIS. This information is transmitted to the Worklist server via HL7 v2.x order messages (ORM), where it acts as a bridge to the radiology devices, ensuring technicians have all necessary information before performing a study.

## 4 Process mining for radiological workflows

This case study aimed to identify improvement opportunities in the radiology department by comparing discovered vs. intended process models to detect deviations, bottlenecks, and exceptions. The central metric was the end-to-end patient journey time, from study request to report availability. This comprehensive view assesses both operational efficiency (e.g., scheduling wait times) and clinical effectiveness (e.g., image acquisition time). While often focusing on the waiting period as a key bottleneck, examining the complete journey ensures no inefficiencies in other phases are overlooked. Full traceability is essential for safe, efficient, and standardized care. We followed the PM<sup>2</sup> methodology [4] as is, aligning institutional objectives and ensuring active stakeholder participation.

### 4.1 Planning

In this phase, radiological workflows selected include emergency studies (without prior coordination, i.e., agenda) and those with an agenda (scheduled appointments). The main research questions identified for the analysis are:

1. Are there differences in patient attention times for emergency studies and those scheduled? And from the request creation to the start of the study?
2. Are the process times in accordance with the institutional criteria defined? 24 hours maximum for emergency studies and one month for scheduled ones.
3. What is the impact of using the worklist on the efficiency of each flow?
4. What is the degree of compliance with the SWF standard, focusing on communication between the systems (HIS/RIS and radiological modalities)?
5. Are there any inconsistencies within the data registered that affect traceability of the studies (e.g., studies without images or associated reports)?

The systems identified are: i) HIS/RIS to obtain the patient records, the appointments (agenda), emergency and requests, ii) the PACS system, which stores images and metadata of the studies (start/end events), and iii) the Worklist, which is the critical interface to send patient and study data from the HIS to the imaging equipment. The stakeholders identified are: the Chief of medical

Imaging who is in charge of the definition of acceptable duration of the workflows and their validation criteria, the Systems Development Responsible (first author of this work), who integrates the IT team and is in charge of the data extraction and the technical validation of the data, and the infrastructure and servers team, who provides the hardware and servers to support the processes.

## 4.2 Extraction

The data extraction corresponds to the period October 2021 to March 2024, and the sources were the MySQL database of the HIS and the PACS server. Radiological process modalities covered were Magnetic Resonance Imaging (MR), Computed Tomography (CT), Mammography (MG), and Radiography (CR). Key events included: request creation, appointment scheduling, uploading to the worklist server, starting/completing the imaging study, and creating medical reports. The extraction process involves executing cross-referenced SQL queries to link patient medical events based on the unique identifier attribute at the institution (Registration Number) and the closeness of the event dates. An external program was developed in PHP to access all databases in read-only mode, as directly executing the aforementioned SQL queries on the database would be too complex. The event log was generated in the XES[8] standard format. The extracted data consisted of 602,011 events in 103,198 cases and 12,713 variants.

## 4.3 Data processing

Data quality issues primarily involved duplicates and incomplete records, particularly cases of radiological studies without corresponding images. As a criterion for validating the maximum time between events, we use 62 days between admission to the ER and the start of the study, and 183 days between scheduling of the appointment and the beginning of the study. Also, we excluded cases where the duration was over a year. We performed two sessions with the Chief of Medical Imaging to show the data and analysis results. One of the most significant challenges was addressing the discrepancies in the creation and completion dates of images from the PACS server database. Not every event was recorded; missing events, such as the patient's arrival at the hospital and/or to the radiology room, were identified. The final processed event log consisted of 174,000 events (26%) in 30,024 cases and 999 variants.

## 4.4 Mining and analysis

The mining and analysis phase activities were carried out with Disco<sup>4</sup>. Figure 2 shows the overall performance models for scheduled and emergency studies.

Regarding workflow disparities (Question 1), it was observed that in the MR modality, there is a low level of integration of the worklists (only 20% of appointments), which generates significant delays in the start of the studies. On

<sup>4</sup> <https://fluxicon.com/disco/>

the other hand, X-rays showed a high dependence on emergency events (69% of cases) but with a minimal report generation rate (only 4% of X-rays), evidencing bottlenecks in the final phase of the process. CT stands out as the most efficient modality, with a median time of only 2.1 hours from the start of the study to the generation of the report, in contrast to other modalities, where this process can extend to days. In the overall analysis (Question 2), times also vary significantly depending on the type of care. In emergencies, the median time from the creation of the order request to the start of the study is 1 hour and 25 minutes (transition with red box in Figure 2a), while for scheduled studies it reaches 11 days (transition with red box in Figure 2b). Although these medians meet institutional criteria, a further analysis of the distribution showed that significant outliers exceeding these timeframes occurred only seldomly.

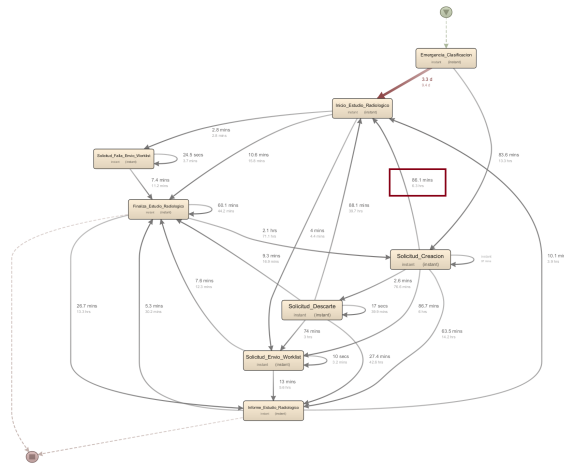
Regarding the use of Worklists (Question 3), 75% of the studies utilize Worklists to some extent, although they may not fully comply with the SWF standard. In the specific case of MRI, for emergency events, it is only used in 38% of cases, which highlights opportunities to improve operations.

It was identified that only 45.2% of cases correctly complied with the SWF standard (Question 4). Noncompliance or partial compliance with the SWF (Standard Workflow) disrupts the correct sequence of procedures in the radiology workflow protocol. Deviations such as generating reports before study completion or accessing the worklist midprocedure can lead to diagnostic errors and loss of patient study traceability. These workflow failures compromise process integrity, increase clinical risks, and may result in misplaced or unlinked data, affecting patient monitoring and follow-up accuracy.

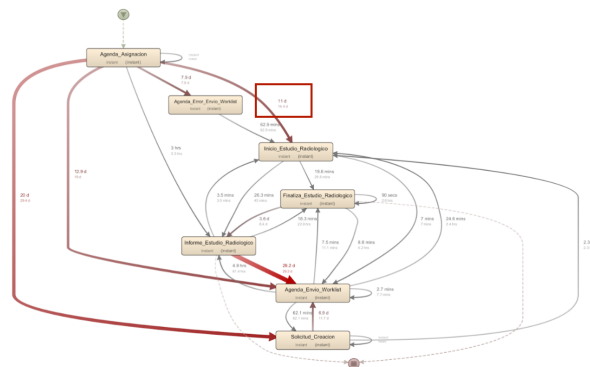
We detected significant inconsistencies (Question 5): 11% of appointments were not linked to completed studies, and 6.69% of emergency requests did not result in the actual start of studies. Our primary hypothesis is that these anomalies occur because healthcare professionals bypass the standardized worklist, creating a critical gap in digital traceability. This manual deviation explains the data discrepancies and poses a significant patient safety risk.

## 4.5 Evaluation

The analysis revealed significant bottlenecks in scheduling times, with medians between 11 and 13 days. While within institutional limits, this delay represents the slowest process stage, causing potential treatment delays, even for severe conditions requiring prompt attention. Under-utilization of MRI worklists was confirmed, impacting efficiency. A critical system error was identified in 700 cases, where multiple patients shared a single appointment due to a recursive "Schedule Assignment" activity, thereby violating the one-appointment-per-patient principle. Comparison of emergency and scheduled events highlighted specific time criteria. Linking to standards reaffirmed the value of time analysis aligned with the SWF. Stakeholder-defined criteria, particularly from the Chief of Medical Imaging, were key in connecting research questions to clinical needs.



(a) Emergency studies



(b) Scheduled studies

Fig. 2: Performance analysis according to the type of process start

## 4.6 Process improvement and Support

Among the recommended improvements are creating automated alerts for studies with pending reports, implementing additional restrictions in the HIS to prevent redundant scheduling, and promoting training opportunities on the worklist for radiology practitioners. A positive impact was the immediate implementation of an improvement that involved correcting a bug in the scheduling module that had gone undetected for 14 years after it went into production. There is also a motivation to promote feedback among stakeholders, primarily by increasing the work on validating findings between clinical teams and IT staff. One of the ac-

tions that should be prioritized is the implementation of controls for mandatory use of the worklist and reviewing scheduling policies.

## 5 Characteristics and challenges

As a second objective, we evaluated the distinctive characteristics and challenges of applying process mining in healthcare, as identified in [10]. Table 1 summarizes the characteristics observed in the case study, the challenges encountered, and the actions that can be taken to address them.

Table 1: Mapping between distinguishing characteristics and challenges observed

| Characteristics manifested                                                   | Related challenges and action taken                                                                               |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| D1: 16 frequent variants over >12,000 studies; large temporal dispersion     | Separate modeling/monitoring per modality and context (C1); go beyond discovery (C2) and monitor drift (C3).      |
| D2: Infrequent behavior, e.g., 700 double-booked agendas and rare MG in ER   | Root-cause analysis and fix the scheduler module (C2); fix data-quality (C6)                                      |
| D3: Multiple deviations, e.g., 15% studies without report, CR outside agenda | Embed SWF-like rules and validations in HIS (C2, C3); HIS must embed a process perspective (C9)                   |
| D4: Execution without Worklist or schedule validation                        | Enforce Worklist usage and add async alerts for rule violations (C4), and process-aware validations (C9)          |
| D5: Need to merge admin/clinical/technical layers                            | Cross-layer reconciliation and data quality checks (C6, C9)                                                       |
| D6: Multidisciplinary correction of an old time bug                          | Shared dashboards and training in basic process mining concepts (C5); sustain data-quality governance (C6)        |
| D7: Missing reports and modality mismatches affect diagnostic continuity     | Business rules needed and MRN-centric tracking (C7, C8)                                                           |
| D8: Trace visualizations exposed incomplete flows                            | Patient-centred (C8) dashboards by modality (C2, C5)                                                              |
| D9: Duplicates, timestamp issues, weak traceability found                    | Permanent data quality policy (C6), refactor scheduling code (C9), and address privacy and security concerns (C7) |
| D10: Legacy HIS under current load and new modalities                        | Real-time alerts, auto-controls, and drift monitoring (C3, C10)                                                   |



### 5.1 Distinguishing Characteristics

We assessed whether the distinguishing characteristics were present in the project context and to what extent they posed a challenge.

- D1: Exhibit Substantial Variability** The process exhibits high structural and temporal variability: 16 frequent variants across more than 12,000 studies. Modality, clinical context (emergency vs. outpatient), the (absence of) medical report, and (non-)use of scheduling reshape the flow. Emergency (ER)-initiated cases without referral resolve within hours, while scheduled cases can take up to 11 days. Radiography (CR), lacking a defined scheduling mechanism and with only 4% of studies having a report, shows the widest dispersion. **Implication:** Heterogeneity renders monolithic modelling ineffective and motivates modality - and context- specific analyses.
- D2: Value the Infrequent Behaviour** We detected infrequent yet critical behaviors: (i) 700 double-booked agendas, (ii) radiographic studies executed via ER despite having been scheduled as another modality, and (iii) 11 Mammography (MG) requests in ER without clear clinical justification. **Implication:** Although infrequent, these behaviors are essential for understanding systemic vulnerabilities and guiding corrective actions.
- D3: Use Guidelines and Protocols** Although SWF was adopted, implementation revealed multiple deviations: CR operates outside the scheduling system, and 15% of studies lack a medical report. Missing automatic validations enabled skipping critical steps. **Implication:** Need to reinforce operational protocols through embedded rules and validations in the HIS.
- D4: Break the Glass** The system permits uncontrolled workflow jumps: execution without schedule validation, studies without Worklist, and agendas shared by multiple patients. **Implication:** Flexibility in ER must be governed by explicit rules and alerts.
- D5: Consider Data at Multiple Abstraction Levels** Traceability required merging administrative (scheduling), clinical (requests/reports), and technical (PACS) layers. Multiple-booking errors emerged from metadata-level repetitions; missing reports or modality mismatches surfaced only after cross-layer integration. **Implication:** PM4H pipelines must integrate heterogeneous layers with robust data quality checks.
- D6: Involve a Multidisciplinary Team** Radiologists, technologists, medical records staff, HIS developers, and infrastructure technicians collaborated to detect and fix a long-standing scheduler bug. **Implication:** Multidisciplinary is essential to interpret deviations and enact realistic improvements.
- D7: Focus on the Patient** Missing reports, modality mismatches, and long delays affect patient safety and diagnostic continuity. The Medical Record Number served as the backbone to reconstruct trajectories. **Implication:** Patient-centric conformance and clinical closure rules are needed.

- D8: Think about White-box Approaches** Trace visualizations exposed incomplete flows, absent reports, and scheduled-executed modality divergences. **Implication:** Targeted technical fixes are required, and clinical-friendly dashboards to monitor modality compliance are needed.
- D9: Generate Sensitive and Low-Quality Data** We found timestamp inconsistencies, duplicated events, weak patient linkage, low Worklist usage, and the scheduler error compromising historical reliability. **Implication:** Immediate corrective measures are prompted, such as implementing record cross-validation protocols, and a permanent data quality monitoring policy.
- D10: Handle Rapid Evolutions and New Paradigms** The HIS was built for a much smaller volume and fewer modalities. CT shows that efficiency and traceability are achievable with best practices. **Implication:** Future upgrades must include real-time alerts, automated controls, and dynamic process analysis to meet the evolving needs of hospitals.

## 5.2 Challenges

We analyzed whether the challenges emerged during the project, how they were addressed, and what insights they provide for guiding future work.

- C1: Design Dedicated/Tailored Methodologies and Frameworks** While PM<sup>2</sup> structured the analysis, modalities showed distinct behaviors: CR lacks scheduling, MG is fully scheduled, and CT combines both approaches. These differences (D1) require distinct models, KPIs, and conformance rules. We have separated analysis per modality and context
- C2: Discover Beyond Discovery** Critical traces were detected, including the repetition of the appointment assignment activity for different patients, which violates basic process rules. Substantial differences were also observed by modality: for instance, CT showed workflows with high Worklist usage and efficient report times. Given the appearance of characteristics (D1, D2, D3, D5, and D8), we complemented discovery with conformance checking and enhancement to expose rule violations and quantify compliance.
- C3: Mind the Concept Drift** The long-standing programming error in appointment assignment, together with unscheduled MG in emergency and low MRI Worklist usage, reflects process evolution not mirrored in the HIS. Heterogeneous pathways (D1), persistent deviations from protocols (D3), and the rapid evolution of practice and technology (D10) call for sliding-window, drift-aware monitoring to keep models up to date.
- C4: Deal with Reality** Real data analysis revealed modality-specific differences: CT showed the fastest report turnaround, whereas MG experienced reporting delays despite using the Worklist. CR studies occurred unscheduled. Uncontrolled bypasses (D4), the necessity to integrate administrative/-clinical/technical layers (D5), and the presence of sensitive, imperfect data (D9) highlight the critical role of real data.

- C5: Do It Yourself (DIY)** Real-time monitoring tools could enable clinical staff to detect modality-specific variations, such as 80% of scheduled MR studies skipping the Worklist. Multidisciplinary collaboration (D6) and interpretable tooling (D8) are key to empowering teams to detect and correct deviations proactively.
- C6: Pay Attention to Data Quality** Modality-specific analysis revealed hidden issues: CR studies lacked request records, while MR traces failed to create requests. Infrequent yet critical behaviours (D2), multi-layer integration requirements (D5), multidisciplinary ownership (D6), and sensitive/low-quality data issues (D9) highlight the need for data quality checks to include modality-specific clinical coherence.
- C7: Take Care of Privacy and Security** Built-in encryption was used for anonymizing patient and staff data. However, low Worklist usage in MRI risks incorrect image-patient linkage, compromising traceability. From a patient-centred view (D7) and given the sensitivity and quality limitations of the data (D9), strengthening the integration between HIS records and execution workflows is essential for both data integrity and care quality.
- C8: Look at the Process through the Patient’s Eyes** Patient-centric perspective shows CT with almost immediate closure vs. MG with significant report delays. Patient-centred analysis (D7), supported by interpretable, white-box visualisations (D8), closes gaps and improves perceived quality.
- C9: Complement HISs with the Process Perspective** The HIS fails to detect multi-patient appointments or CR studies in incorrect workflows, nor does it flag MRI Worklist underuse. Enforcing protocol conformance (D3), constraining uncontrolled behaviours (D4), integrating multi-level data (D5), and managing sensitive/low-quality data (D9) require embedding process-aware validation rules and automated alerts directly into the HIS.
- C10: Evolve in Symbiosis with the Developments in the HC Domain** Process mining enabled structural improvements: reinforcing CT’s efficient Worklist/reporting practices, restructuring MG workflows to accelerate reporting, and mandating Worklist use and process revisions in MRI. Protocol-governed workflows (D3) and the domain’s rapid evolution (D10) imply that PM4H must co-evolve with clinical practice, supported by tools ready to integrate additional data sources for continuous, evidence-based optimization.

## 6 Conclusions

In this paper, we performed a case study applying process mining to radiological workflows at the Hospital de Clínicas. The project revealed critical problems that need to be addressed and led to immediate system-level fixes (e.g., scheduler refactoring, Worklist enforcement). We also provide evidence that PM4H characteristics and challenges manifest in this context, and that PM4H becomes a practical instrument that moves beyond retrospective description by enabling verifiable, system-level improvements in radiology workflows.

We acknowledge construct, internal, and external validity threats: the case study is single-site; the event log required non-trivial reconciliation across layers; and some KPIs were institution-specific, limiting generalization.

Future work is needed to operationalise the insights, e.g., formal conformance quantification against SWF, automated concept drift detection, embedding real-time alerts and dashboards for DIY monitoring by clinical teams, and incorporating process-aware rules directly into the HIS to sustain continuous compliance and data-quality governance. Moreover, future PM4H initiatives should prospectively evaluate the impact of the operational changes introduced.

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